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Detonator for a perforating gun assembly

Abstract

A detonator for a perforating gun. The detonator is used to ignite a detonator cord in a charge tube for a perforating gun assembly. The detonator includes a cartridge. The cartridge comprises a tubular body having a first end and a second end. Preferably, the tubular body is fabricated from a metallic material. The detonator also comprises a post proximate the first end of the tubular body. The post is configured to be in electrical communication with a first leg wire connected to an addressable switch. Alternatively, the post is in contact with the end of a detonation pin or with a pair of hot terminals. The detonator also has a fuse head within the bore of the tubular body, and a fuse wire. The fuse head is in contact with the explosive charge material. The detonator is releasably snapped into a compartment located within the charge tube.

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Background/Summary

STATEMENT OF RELATED APPLICATIONS (1) The present application claims the benefit of U.S. Ser. No. 63/508,985 filed Jun. 19, 2023. That application was titled “Detonator for a Perforating Gun Assembly.” (2) The present application also claims the benefit of U.S. Ser. No. 63/384,474 filed Nov. 21, 2022. That application was titled “Detonator for a Perforating Gun Assembly.” (3) This application is also filed as a Continuation-in-Part of U.S. Ser. No. 17/543,121 (1312.0007-US5) filed Dec. 6, 2021. That application was titled “End Plate For A Perforating Gun Assembly.” (4) The '121 application was filed as a Divisional of U.S. Ser. No. 17/175,651 (1312.0007-US3) filed Feb. 13, 2021. That application was titled “Detonation System Having Sealed Explosive Initiation Assembly.” The '651 application issued on Apr. 5, 2022 as U.S. Pat. No. 11,293,737. (5) The '651 application was filed as a Continuation-in-Part of U.S. Ser. No. 16/996,692 filed Aug. 18, 2020 (1312.0007-US2). That application is also entitled “Detonation System Having Sealed Explosive Initiation Assembly.” The '692 application issued on Aug. 2, 2022 as U.S. Pat. No. 11,402,190. (6) Each of these applications is incorporated herein in its entirety by reference.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT
(1) Not applicable.
THE NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT
(2) Not applicable.
BACKGROUND OF THE INVENTION
(3) This section is intended to introduce various aspects of the art, which may be associated with exemplary embodiments of the present disclosure. This discussion is believed to assist in providing a framework to facilitate a better understanding of particular aspects of the present disclosure. Accordingly, it should be understood that this section should be read in this light, and not necessarily as admissions of prior art.
TECHNICAL FIELD OF THE INVENTION
(4) The present disclosure relates to the field of hydrocarbon recovery operations. More

specifically, the present subject matter relates to a perforating gun assembly used for the perforation of steel casing in a wellbore. Further still, the present invention relates to a detonator used to ignite a detonator cord in a perforating gun charge tube.

DISCUSSION OF THE BACKGROUND

(5) For purposes of this disclosure, U.S. Pat. No. 11,402,190 will be referred to as “the parent application.” The parent application has been incorporated herein in its entirety by reference.

(6) In the drilling of an oil and gas well, a near-vertical wellbore is formed through the earth using a drill bit urged downwardly at a lower end of a drill string. After drilling to a predetermined depth, the drill string and bit are removed and the wellbore is lined with a string of casing. An annular region is thus formed between the string of casing and the formation penetrated by the wellbore.

(7) A cementing operation is conducted in order to fill or “squeeze” the annular area with cement along part or all of the length of the wellbore. The combination of cement and casing strengthens the wellbore and facilitates the zonal isolation, and subsequent completion, of hydrocarbon-producing pay zones behind the casing.

(8) In connection with the completion of the wellbore, several strings of casing having progressively smaller outer diameters will be cemented into the wellbore. These will include a string of surface casing, one or more strings of intermediate casing, and finally a string of production casing. The process of drilling and then cementing progressively smaller strings of casing is repeated until the well has reached total depth or “TD.” In some instances, the final string of casing is a liner, that is, a string of casing that is not tied back to the surface.

(9) Within the last two decades, advances in drilling technology have enabled oil and gas operators to “kick-off” and steer wellbore trajectories from a vertical orientation to a near-horizontal orientation. The horizontal “leg” of each of these wellbores now often exceeds a length of one mile, and sometimes two or even three miles. This significantly multiplies the wellbore exposure to a target hydrocarbon-bearing formation. The horizontal leg will typically include the production casing.

(10) FIG. **1** is a side, cross-sectional view of a wellbore **100**, in one embodiment. The wellbore **100** has been completed horizontally, that is, it is completed with a horizontal leg **156**. The wellbore **100** defines a bore **10** that has been drilled from an earth surface **105** into a subsurface **110**. The wellbore **100** is formed using any known drilling mechanism, but preferably using a land-based rig or an offshore drilling rig on a platform.

(11) The wellbore **100** is completed with a first string of casing **120**, sometimes referred to as a surface casing. The wellbore **100** is further completed with a second string of casing **130**, typically referred to as an intermediate casing. In deeper wells, that is, wells completed below 7,500 feet, at least two intermediate strings of casing will be used. In FIG. **1**, a second intermediate string of casing is shown at **140**.

(12) The wellbore **100** is finally completed with a string of production casing **150**. In the view of FIG. **1**, the production casing **150** extends from the surface **105** down to a subsurface formation, or “pay zone” **115**. As noted, the wellbore **100** is completed horizontally, meaning that a near-horizontal section or “leg” **156** is provided. The production casing **150** extends substantially across the horizontal leg **156**.

(13) It is observed that an annular region around the surface casing **120** is filled with cement **125**. The cement (or cement matrix) **125** serves to isolate the wellbore **100** from fresh water zones and potentially porous formations around the surface casing **120**.

(14) The annular regions around the intermediate casing strings **130**, **140** are also filled with cement **135**, **145**. Similarly, the annular region around the production casing **150** is filled with cement **155**. However, the cement **135**, **145**, **155** is placed behind the respective strings of casings **130**, **140**, **150** up to the lowest joint of the immediately surrounding casing string. Thus, a non-cemented annular region **132** is typically preserved above the cement matrix **135**, a non-cemented annular region **142** may optionally be preserved above the cement matrix **135**, and a non-cemented

annular region **152** is frequently preserved above the cement matrix **155**.

(15) The horizontal leg **156** of the wellbore **100** includes a heel **153** and a toe **154**. In this instance, the toe **154** defines the end of the wellbore **100** at total depth (or "TD"). In order to enhance the recovery of hydrocarbons, particularly in low-permeability formations, the production casing **150** along the horizontal section **156** undergoes a process of perforating and fracturing. Due to the exceptionally long lengths of new horizontal wells, the perforating and formation treatment process is carried out in stages.

(16) In one method, a perforating gun assembly **200** is pumped down the wellbore **100** towards the toe **154** at the end of a wireline **240**. The perforating gun assembly **200** will include a series of perforating guns (shown at **210** in FIG. **2**), with each perforating gun **210** having sets of charges ready for detonation. The charges associated with one of the perforating guns **210** are detonated, and perforations (not shown) are "shot" into the production casing **150**. Those of ordinary skill in the art will understand that perforating guns have explosive charges, typically shaped, hollow, or projectile charges, which are ignited to create holes in the casing (and, if present, the surrounding cement) **150** and to pass at least a few inches and possibly several feet into the formation **115**. The perforations create fluid communication with the surrounding formation **115** (or pay zone) so that hydrocarbon fluids can flow into the casing **150**.

(17) After perforating, the operator will fracture the formation **115** through the perforations. This is done by pumping treatment fluids into the formation **115** at a pressure above a formation parting pressure. Those of ordinary skill in the art will understand that "formation parting pressure" is a reference to the downhole pressure required to open up the rock formation to receive fluids. This is in contrast to the hydraulic fracturing pressure, or pumping pressure, measured at the surface in psig.

(18) After the fracturing operation is complete, the wireline **240** will be raised within the casing **150** from the surface **105**, and the perforating gun assembly **200** will be positioned at a new location (or "depth") along the horizontal wellbore **156**. A plug, such as plug **112**, is set below the perforating gun assembly **200** using a setting tool **160**, and new shots are fired in order to create a new set of perforations. Thereafter, treatment fluid is again pumped into the wellbore **100** and into the formation **115**. In this way, a second set (or "cluster") of fractures is formed away from the horizontal leg **156** of the wellbore **100**.

(19) The process of setting a plug, perforating the casing, and fracturing the formation is repeated in multiple stages until the wellbore **100** has been completed, that is, it is ready for production. In FIG. **1**, it can be seen that two separate plugs **112** have been placed along the horizontal leg **156** of the wellbore. Of course, it is understood that the completed horizontal leg **156** of the wellbore **100** may extend many hundreds of feet, with multiple plugs **112** being set between the stages. The plugs **112** may be at 150 foot to 250 foot spacing. A string of production tubing (not shown) is then placed in the wellbore **100** to provide a conduit for production fluids to flow up to the surface **105**.

(20) In order to provide perforations for the multiple stages without having to pull the perforating gun assembly **200** after each detonation, the perforating gun assembly **200** employs multiple guns in series. FIG. **2** is a side view of an illustrative perforating gun assembly **200**, or at least a portion of an assembly. The perforating gun assembly **200** comprises a string of individual perforating guns **210**.

(21) Each perforating gun **210** represents various components. These typically include a "gun barrel" **212** which serves as an outer tubular housing. An uppermost gun barrel (or "gun barrel housing") **212** is supported by an electric wire (or "e-line") **240** that extends from the surface **105** and delivers electrical energy down to the perforating gun assembly **200**, also known as a "tool string." Each perforating gun **210** also includes an explosive initiator, or "detonator" (shown in phantom at **229**). The detonator **229** is typically a small aluminum housing having an internal resistor. The detonator **229** receives electrical energy from the surface **105** and through the e-line **240**, which heats the resistor.

(22) In a typical perforating gun **210**, the detonator **229** is a so-called block detonator. This is because the detonator **229** is held in place adjacent a detonator cord by means of a non-conductive block. The detonator cord passes through the block, with the detonator **229** residing at least partially inside of the block adjacent the detonator cord.
(23) In practice, a pair of wires (referred to as “leg wires”) extend from an addressable switch to the detonator **229**. These wires are frequently connected once the perforating guns **210** are delivered to a well site. This means the wires must be accurately and safely connected in conditions that are sometimes hostile, e.g., conditions of blowing sand, rain, snow, or extreme cold.
(24) Resistors may be connected to bridge wires within the block. When the wires are energized downhole, electrical current passes through the resistors and to a bridge wire, causing the bridge wire to become heated. The bridge wire is a so-called hot voltage wire. The bridge wire is in proximity to a chamber holding an explosive material, such as a nitroamine. The most common explosive is an organic chemical compound known as RDX. This may be referred to as a base charge.
(25) The detonator **229** is in close proximity to a detonator cord. The detonator cord may represent a poly-braid material that holds the RDX explosive, with a nylon or aluminum jacket surrounding the poly-braid material. When the base charge within the detonator **1000** is heated, a small explosion is set off that melts the jackets of the detonator cord and ignites the explosive compound residing therein.
(26) When ignited, the detonator cord initiates one or more shots, typically referred to as “shaped charges.” The shaped charges (one shown at **320** in FIG. **3**) are held in a charge tube (shown at **300** in FIG. **3**) and discharge through openings **312** in the charge tube **300** and opening **215** in the selected gun barrel **212**. As the RDX, or other suitable explosive, is ignited, the detonator cord propagates an explosion down its length to each of the charges **320** along the charge tube **300**.
(27) The perforating gun assembly **200** may also include short centralizer subs **220**. The assembly **200** also includes the charge tubes **300**, which reside within the gun barrel housings **212** and are not visible in FIG. **2**. In addition, tandem subs **225** are used to connect the gun barrel housings **212** end-to-end. Each tandem sub **225** comprises a metal threaded connector placed between the gun barrels **210**. (A complete tandem sub is shown at **400** in FIG. **4** of the parent application.) Typically, the gun barrels **210** will have female-by-female threaded ends while the tandem sub **225** has opposing male threaded ends (indicated at **404** in FIG. **4** of the parent application).
(28) The perforating gun assembly **200** with its long string of gun barrels, which include the housings **212** of the perforating guns **210** and the charge tubes **300**, is carefully assembled at the surface **105**, and then lowered into the wellbore **100** at the end of the e-line **240**. The e-line **240** extends upward to a control interface (not shown) located at the surface **105**. An insulated connection member **230** connects the e-line **240** to the uppermost perforating gun **210**. Once the assembly **200** is in place within the wellbore, an operator of the control interface sends electrical signals to the perforating gun assembly **200** for detonating the shaped charges **320** and for creating perforations into the casing **150**.
(29) As noted in FIG. **1**, a setting tool **160** resides at the end of the perforating gun assembly **200**. The setting tool **160** may be connected to the lowermost perforating gun **210** by means of a tandem sub **225**. The setting tool **160** is used to set the plug **112** along the wellbore **100** at a desired depth. This is typically done by using an igniter which initiates the burning of a power charge.
(30) After the casing **150** has been perforated and at least one plug **112** has been set, the setting tool **160** and the perforating gun assembly **200** are removed from the wellbore **100** and a ball (not shown) is dropped into the wellbore **100** to close the plug **112**. When the plug **112** is closed, a fluid (e.g., water, water and sand, fracturing fluid, etc . . .) is pumped by a pumping system down the wellbore **100** (typically through coiled tubing) and through the newly-formed perforations for fracturing purposes. For a formation fracturing operation, the pump rate will create downhole pressure that is above the formation parting pressure.

(31) As noted, the above operations may be repeated multiple times for perforating and/or fracturing the casing **150** at multiple depths, corresponding to different stages of the wellbore **100**. Multiple plugs **112** may be used for isolating the respective stages from each other during the fracturing phases. When all stages are completed, the plugs **112** are drilled out and the wellbore **100** is cleaned using a circulating tool.
(32) It can be appreciated that a reliable signal must be provided to the detonator to ensure that the charges along the gun barrel are detonated. Currently, detonators are manufactured by filling a small, extruded aluminum tube with explosive material and then crimping a fuse head assembly in place. In this case, the fuse head assembly comprises the two leg wires, the resistors, the bridge wire (forming an ignition point), and a grommet that holds all components in place. Current detonators require wiring and connecting the wire legs by hand. Those of ordinary skill in the art will understand that the wires themselves are a fail point as they can easily break off or pull out of the grommet along the detonator.
(33) Accordingly, a need exists for a detonator that can be easily assembled without need of soldering wires together in the shop or in the field. A need further exists for a detonator that may be pre-wired with the two leg wires of the addressable switch and then dropped into place in a charge tube, ready for run-in into a wellbore. A need further exists for a detonator wherein the detonator may simply be snapped into a compartment along the charge tube of a perforating gun assembly at a well site, thereby placing the detonator in electrical communication with the addressable switch without need of manipulating wires in the field.
SUMMARY OF THE INVENTION
(34) A detonator for a perforating gun is provided. The detonator is used to ignite a detonator cord within a perforating gun assembly. The perforating gun assembly, in turn, is used for perforating a wellbore. The perforating gun assembly has a charge tube holding a plurality of charges, and a gun barrel housing holding the charge tube. The perforating gun assembly also includes an addressable switch which interprets signals sent from the surface.
(35) Beneficially, for the present invention the charge tube has a compartment. The compartment is used to hold the detonator, preferably through a snap-fit or friction fit type arrangement. In this instance, the detonator is in the form of a cartridge. The cartridge may be inserted into the compartment in the field.
(36) The detonator first comprises a tubular body. The tubular body has a first end, and a second end opposite the first end. Preferably, the tubular body is fabricated from a metallic material such as aluminum. The detonator is in the form of a cartridge.
(37) The detonator also comprises a terminal. The terminal resides at the first end of the tubular body and is fabricated from a conductive material. The terminal is configured to be in electrical communication with the addressable switch along the perforating gun assembly. In one aspect, electrical communication is accomplished through a first leg wire extending from the addressable switch. More preferably, electrical communication is obtained by inserting the cartridge into the compartment such that the terminal is in contact with the end of a detonation pin. The detonation pin transmits a detonation signal to the detonator within the compartment.
(38) The terminal may comprise a spring. More preferably, the terminal defines a post. In either instance, the terminal is fabricated from a conductive material such as copper or brass.
(39) In one aspect, the detonator has an upper bore which extends through the post. The upper bore receives a resistor wire. In another aspect, an end of the first leg wire extends into and is connected to an inner wall of the post. In this instance, the resistor wire and the first leg wire may be the same wire.
(40) A lower bore is also preserved along the tubular body. The lower bore houses an explosive charge material below the post.
(41) The detonator also has an initiator. The initiator resides at least partially within the cartridge and is configured to be placed in electrical communication with the addressable switch. This may

be done either by means of the first leg wire coming off of the addressable switch, or by contact with a detonation pin. The initiator comprises a resistor. The resistor is in electrical communication with the terminal by means of the resistor wire.
(42) The initiator further includes a ground wire. The ground wire extends from the resistor and is connected to an internal wall of the tubular body.
(43) As noted, the cartridge is designed to be inserted into a compartment within the charge tube. The compartment contains a ground terminal. In one aspect, the ground terminal is in electrical communication with a second leg wire extending from the addressable switch. In this instance, the ground wire is in electrical communication with the second leg wire through the ground terminal and through the tubular body itself. Inserting the cartridge into the compartment places the tubular body in contact with the ground terminal.
(44) In another instance, the second leg wire of the addressable switch is wrapped around or crimped to the tubular body adjacent the ground wire to provide grounding.
(45) In a preferred arrangement, an insulative element is provided as part of the detonator. The insulative element separates the post at the upper end of the cartridge from the lower bore within the tubular body. The insulative material may be fabricated from molded rubber or synthetic rubber or other non-conductive material. In one aspect, the resistor wire extends through the insulative element and down to the resistor. In this instance, the resistor is in contact with the explosive material within the lower bore of the tubular body.
(46) In an alternate embodiment, the initiator comprises two resistors, indicated as a first resistor and a second resistor. The first resistor resides along the insulative element while the second resistor resides within or below the insulative element but above the explosive material. In addition, the initiator includes a catalyst element residing between the first resistor and the second resistor. The catalyst element is placed within the lower bore and is in contact with the explosive material. A fuse wire resides between the first resistor and the second resistor, with the catalyst element residing along the fuse wire.
(47) Optionally, the insulative element further comprises a flange. The flange is located between upper and lower ends of the insulative element. The flange is configured to seat within a compartment of the perforating gun charge tube. In an arrangement, an end of the first leg wire is wrapped around the upper end of the insulative element while an end of the ground wire is wrapped around the lower end of the insulative element. The upper end of the insulative element (with the first leg wire wrapped around it) may be pushed up into the cap, and then the lower end of the insulative element (with the second leg wire wrapped around it) may be pushed down into the tubular body. In this way the wires cannot be pulled from the detonator during operation.
(48) Preferably, the detonator resides within the perforating gun charge tube adjacent a detonator cord. The uniquely configured compartment provides a friction fit or a snap-fit type arrangement for the detonator adjacent the detonator cord. This may be done, for example, through clips. The detonator is snapped into the receptacle within the charge tube by accessing an opening provided along the charge tube.
(49) In a preferred arrangement, the perforating gun is adjacent a carrier end plate. The carrier end plate holds a detonation pin which is in electrical communication with the addressable switch. As noted above, placement of the cartridge into the compartment causes the terminal to be in physical contact with the addressable switch through the detonation pin.
(50) A method of firing charges into a wellbore casing is also provided herein. The wellbore casing resides within the horizontal portion of a wellbore.
(51) In one embodiment, the method first comprises providing a perforating gun assembly. The perforating gun assembly has a gun barrel housing, and a charge tube residing within the gun barrel housing. A plurality of charges reside along the charge tube, with a detonator cord extending to each of the charges within the charge tube. The perforating gun assembly also has an addressable switch.

(52) The method also includes providing a detonator. The detonator comprises a tubular body. The tubular body has an upper end and a lower end. The tubular body is fabricated from a conductive metal such as aluminum and serves as a cartridge.
(53) The detonator also includes a terminal. The terminal resides at the upper end of the tubular body. The terminal is also fabricated from a conductive metal, such as brass. The terminal may be a spring, or it may be a post biased upwardly by a spring.
(54) The detonator also provides a lower bore. The lower bore resides within the tubular body. The lower bore houses an explosive charge material proximate the lower end of the tubular body.
(55) An initiator also resides within the tubular body. The initiator resides at least partially within the lower bore and is in contact with the explosive material. As noted above, the initiator will include a resistor wire, a resistor, and a ground wire.
(56) The method also includes placing the detonator into a compartment within the charge tube. In this step, the terminal is configured to be placed in electrical communication with the addressable switch. The compartment comprises a ground terminal. Placement of the detonator into the compartment places the tubular body in contact with the ground terminal. Preferably, the detonator is placed into the compartment using a friction-fit or snap-fit arrangement.
(57) In one arrangement, the terminal is in electrical communication with the resistor by means of a resistor wire. At the same time, the perforating gun is adjacent a carrier end plate. The carrier end plate comprises a detonation pin in electrical communication with the addressable switch. Placement of the cartridge into the compartment causes the terminal to be in contact with the addressable switch through the detonation pin.
(58) The method may further comprise: pumping the perforating gun assembly into the horizontal portion of the wellbore using an electric wireline; sending an electrical signal from a surface and down the electric wireline to the perforating gun assembly, wherein the signal reaches the addressable switch; and sending an initiation signal from the addressable switch, through the detonation pin, and to the detonator, causing the detonator cord to be ignited and causing the plurality of charges to fire into the wellbore casing.
(59) A separate method is also provided herein, which is a method of preparing a perforating gun assembly. A first step in this method is providing a perforating gun assembly. The perforating gun assembly has a gun barrel housing, a charge tube residing within the gun barrel housing, a plurality of charges residing along the charge tube, and a detonator cord extending to each of the charges within the charge tube. The perforating gun assembly also has an addressable switch, a detonation pin in electrical communication with the addressable switch, and a compartment within the charge tube.
(60) The method also includes providing a detonator. The detonator comprises: a tubular body having an upper end and a lower end; a terminal residing at the upper end; a lower bore within the tubular body housing an explosive charge material proximate the lower end; and an initiator in contact with the explosive material.
(61) The method also includes transporting the perforating gun assembly to a well site. After the perforating gun assembly has arrived at the well site, the method comprises placing the detonator into the compartment within the charge tube. In this way, the terminal is in physical contact with an end of the detonation pin.
(62) Preferably, the detonator is placed into the compartment using a friction-fit or snap-fit arrangement.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) So that the manner in which the present inventions can be better understood, certain

illustrations, charts, and/or flow charts are appended hereto. It is to be noted, however, that the drawings illustrate only selected embodiments of the inventions and are therefore not to be considered limiting of scope, for the present subject matter may admit to other equally effective embodiments and applications.

(2) FIG. **1** is a cross-sectional side view of a wellbore. The wellbore is being completed with a horizontal leg. A perforating gun assembly is shown having been pumped into the horizontal leg at the end of a wireline.

(3) FIG. **2** is a side view of a perforating gun assembly. The perforating gun assembly represents a series of perforating guns having been threadedly connected end-to-end. Tandem subs are shown between gun barrels of the perforating guns, providing the threaded connections.

(4) FIG. **3** is a perspective view of an illustrative charge tube for a perforating gun. A charge is shown in separated relation.

(5) FIG. **4**A is a perspective view of the charge tube of FIG. **3**. The charge tube has received a top end plate and a bottom end plate. An electric line is shown extending through the charge tube and to the bottom end plate.

(6) FIG. **4**B is a first side view of the charge tube of FIG. **4**A.

(7) FIG. **4**C is a second side view of the charge tube of FIG. **4**A, opposite the view of FIG. **4**B.

(8) FIG. **4**D is a perspective view of the charge tube of FIG. **4**A. Here, the charge tube is being slidably received from above within a gun barrel housing.

(9) FIG. **5**A is a first perspective view of the bottom end plate of FIG. **4**A. The bottom end plate is connected to the charge tube. Three electrical pins are shown extending out of the end plate.

(10) FIG. **5**B is a second perspective view of the bottom end plate of FIG. **4**A. The charge tube has been removed for illustrative purposes.

(11) FIG. **6**A is a first perspective view of a bridged bulkhead assembly. The bulkhead assembly holds a signal transmission pin and a separate detonation pin.

(12) FIG. **6**B is a second perspective view of the bridged bulkhead assembly of FIG. **6**A. Here, the view is seen from an end opposite that of FIG. **6**A.

(13) FIG. **7** is a cross-sectional view of the bridged bulkhead assembly of FIGS. **6**A and **6**B.

(14) FIG. **8** is a side, cross-sectional view of an explosive initiation assembly. The explosive initiation assembly is threadedly connected at opposing ends to gun barrel housings, forming a portion of a perforating gun assembly. The explosive initiation assembly includes, among other components, a tandem sub, a switch housing, and an addressable switch.

(15) FIG. **9** is a perspective view of a detonator of the present invention, in one embodiment. Here, the detonator is placed within a receptacle of the charge tube for a perforating gun.

(16) FIG. **10** is a perspective side view of the detonator of FIG. **9**.

(17) FIG. **11**A is a side, plan view of the detonator of FIG. **10**.

(18) FIG. **11**B is a cross-sectional view of the detonator of FIG. **10**. The view is taken across Line A-A of FIG. **11**A.

(19) FIG. **12**A is a perspective view of the detonator of FIG. **10**. Here, a spring is shown residing at an upper end of a cartridge.

(20) FIG. **12**B is a cross-sectional view of the detonator of FIG. **12**A.

(21) FIG. **13**A is a cross-sectional view of the detonator of FIG. **10**. Here, the spring at the upper end of the cartridge is visible.

(22) FIG. **13**B is a cut-away view of the detonator of FIG. **13**A. A post has been installed over the spring.

(23) FIG. **14**A is a plan view of a detonator in an alternate embodiment.

(24) FIG. **14**B is a cross-sectional view of the detonator of FIG. **14**A, taken across line A-A of FIG. **14**A.

(25) FIG. **15** is another perspective view of the charge tube of FIG. **9**. The detonator is more clearly seen residing with a compartment of the charge tube.

(26) FIG. **16** is a flow chart demonstrating steps for a method of firing shots into a wellbore casing

DEFINITIONS

(27) For purposes of the present application, it will be understood that the term “hydrocarbon” refers to an organic compound that includes primarily, if not exclusively, the elements hydrogen and carbon. Hydrocarbons may also include other elements, such as, but not limited to, halogens, metallic elements, nitrogen, carbon dioxide, and/or sulfuric components such as hydrogen sulfide.

(28) As used herein, the terms “produced fluids,” “reservoir fluids,” and “production fluids” refer to liquids and/or gases removed from a subsurface formation, including, for example, an organic-rich rock formation. Produced fluids may include both hydrocarbon fluids and non-hydrocarbon fluids. Production fluids may include, but are not limited to, oil, natural gas, pyrolyzed shale oil, synthesis gas, a pyrolysis product of coal, nitrogen, carbon dioxide, hydrogen sulfide, and water.

(29) As used herein, the term “fluid” refers to gases, liquids, and combinations of gases and liquids, as well as to combinations of gases and solids, combinations of liquids and solids, and combinations of gases, liquids, and solids.

(30) As used herein, the term “subsurface” refers to geologic strata occurring below the surface of the earth.

(31) As used herein, the terms “wireline,” “cord,” and “e-line,” when referring to an electrical line, may be used interchangeably to describe electrically conductive cabling used to transmit electrical signals in a wellbore.

(32) As used herein, the term “formation” refers to any definable subsurface region regardless of size. The formation may contain one or more hydrocarbon-containing layers, one or more non-hydrocarbon containing layers, an overburden, and/or an underburden of any geologic formation. A formation can refer to a single set of related geologic strata of a specific rock type, or to a set of geologic strata of different rock types that contribute to or are encountered in, for example: (i) creation, generation, and/or entrapment of hydrocarbons or minerals; and (ii) execution of processes used to extract hydrocarbons or minerals from the subsurface region.

(33) As used herein, the term “wellbore” refers to a hole in the subsurface made by drilling or insertion of a conduit into the subsurface. A wellbore may have a substantially circular cross-section, or other cross-sectional shapes. The term “well,” when referring to an opening in the formation, may be used interchangeably with the term “wellbore.”

(34) Reference herein to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in connection with an embodiment is included in at least one embodiment of the subject matter disclosed. Thus, the appearance of the phrases “in one embodiment” or “in an embodiment” in various places throughout the specification is not necessarily referring to the same embodiment.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

(35) The following description of the embodiments refers to the accompanying drawings. The same reference numbers in different drawings identify the same or similar elements. The following detailed description does not limit the presently disclosed subject matter; instead, the scope of the embodiments herein is defined by the appended claims.

(36) In the following, the terms “upstream” and “downstream” are used to indicate that one gun barrel of a perforating gun may be situated above and one below, respectively. However, one skilled in the art would understand that the presently disclosed subject matter is not limited only to the upstream gun or only to the downstream gun, but in fact, may be applied to either gun. In other words, the terms “upstream” and “downstream” are not necessarily used in a restrictive manner, but only to indicate, in a specific embodiment, the relative positions of perforating guns or other components.

(37) FIG. **3** is a perspective view of an illustrative charge tube **300** for a perforating gun **210**. The charge tube **300** defines an elongated tubular body **310** having a first end **302** and a second end **304** opposite the first end **302**. The charge tube **300** has an inner bore **305** dimensioned to receive

charges **320**. A single illustrative charge is shown at **320** in exploded-apart relation to the charge tube **300**. However, it is understood that the charge tube **300** will typically include several charges **320**.

(38) Openings **312** are provided for receiving the charges **320**. The openings **312** enable the charges **320** to penetrate a surrounding casing string, such as production casing **150** of FIG. **1**, upon detonation.

(39) FIG. **4**A is a perspective view of the charge tube **300** of FIG. **3**. In this view, a pair of end plates have been threadedly connected to opposing ends of the charge tube **300**, forming a part of a perforating gun assembly **400**. These represent a top end plate **420** connected at end **302**, and a bottom end plate **430** connected at the bottom end **304**. The end plates **420**, **430** serve to mechanically enclose the top **302** and bottom **304** ends of the charge tube **300**, respectively.

(40) The end plates **420**, **430** help center the charge tube **300** and its charges **320** within an outer gun barrel (not shown in FIG. **4**A but shown at **210** in FIG. **2**). For this reason, the end plates **420**, **430** may also be referred to as “carrier plates” **420**, **430**. The end plates **420**, **430**, along with a bulkhead **475**, also serve to provide a pressure seal from the explosive charges being detonated in a wellbore, thereby preserving the electronic components residing within adjoining tandem subs **225**.

(41) It is understood that each opening **312** along the tubular body **310** of the charge tube **300** will receive and accommodate a respective charge **320**. Herein, charge **320** may optionally be referred to as a shaped charge or an explosive charge. Each shaped charge **320**, in turn, is designed to detonate in response to an explosive initiation signal (“IE”) passed through a detonator wire (such as detonator wire **540** of FIG. **6**A or a first leg wire coming off of an addressable switch). It is understood that the charge tube **300** and the shaped charge **320** of FIGS. **3** and **4**A are illustrative, and that the presently disclosed embodiments described here within are not limited to any particular type, model, or configuration of charges, charge tubes, or gun barrels unless expressly so provided in the claims.

(42) An electronic detonator (shown schematically at **229** in FIG. **2**) and the detonator cord reside inside the charge tube **300**. The charge tube **300** and the gun barrel **210** are intended together to be illustrative of any perforating gun, with the understanding that the perforating gun assembly provides for a detonator and detonator cord internal to the charge tube **300**.

(43) Extending up from the top end plate **420** is the bulkhead **475**. The bulkhead **475** encloses a contact pin (shown at **470** in FIG. **8**). An end of the contact pin **470** is indicated at **472**. The contact pin **470** is configured to transmit detonation and communication signals from the surface **105** down to addressable switches (not shown) along the gun string **200**.

(44) A signal line **410** is seen extending down from the contact pin **470** and through the charge tube **300**. The signal line **410** further extends through the bottom end plate **430** and down to a next perforating gun (not shown). A signal carried by the signal line **410** is transmitted through a signal transmission pin **720′**. An earlier embodiment of the signal transmission pin **720′** is discussed in greater detail in FIGS. **7**A, **22**A and **22**B of the parent application.

(45) At an opposing end of the charge tube **300**, the bottom end plate **430** is shown. The bottom end plate **430** has a closed end surface (shown at **435** in FIGS. **5**A and **5**B). Three pins are shown extending out of the closed end surface **435**. These represent a ground pin **710** and two electrical pins **720′**, **720″**. In one aspect, ground pin **710** connects to the bottom end plate **430** as an electrical ground, while electrical pins **720′**, **720″** connect to white and green wires, respectively. Electrical pin **720′** serves as a signal transmission pin while electrical pin **720″** serves as a detonation pin.

(46) Details concerning the ground pin **710** are discussed in connection with FIGS. **9**A and **9**B of the parent application and need not be repeated herein. For reference, ground pin **710** is seen in the cross-sectional view of FIG. **8** herein. Note that the ground pin **710** does not extend through the end plate **430** but simply screws into the end surface **435**.

(47) FIG. **4**B is a first side view of the charge tube **300** of FIG. **4**A. FIG. **4**C is a second side view of the charge tube **300** of FIG. **4**A, opposite the view of FIG. **4**B. Of interest, the charges **320** have

been removed, leaving the signal transmission line **410** visible.
(48) FIG. **4**D is another perspective view of the charge tube **300** of FIG. **3**. Here, the charge tube **300** is being slidably received from above within a gun barrel housing **510**. The gun barrel housing **510** has an upper end **502** and a lower end **504**. The gun barrel housing **510** has a length that is generally conterminous with a length of the charge tube **300**. The gun barrel housing **510** includes openings **512** that align with openings **312** of the tubular body **310** when the gun barrel housing **510** is slid in place over the charge tube **300**.
(49) In the view of FIG. **4**D, the gun barrel housing **510** is shown in phantom when placed over the charge tube **300**. The upper end of the gun barrel housing **510** is indicated at **502′** while the lower end is shown at **504′**. Openings, also referred to as scallops along the gun barrel housing **510**, are provided at **512′**. It is understood that a joining of the charge tube **300** and the gun barrel housing **510** typically takes place at the shop before delivery of a perforating gun assembly **400** to a well site.
(50) In the arrangement of FIGS. **4**A through **4**D, the tubular body **310** and gun barrel housing **510** are shown downstream from the contact pin **470**. However, it is understood that a separate charge tube **300** and gun barrel housing **510** reside upstream from the contact pin **470**. Similarly, separate charge tubes **300** and gun barrel housings **510** reside downstream from the pins **710**, **720′**, **720″**, forming what may be a long series of perforating guns **210** in a perforating gun assembly **400**.
(51) FIG. **5**A is a first perspective view of the bottom end plate **430** of FIG. **4**A. The end plate **430** is slidably connected to the tubular body **310** of the charge tube **300** at end **304**. A bolt **810** threadedly connects a proximal end **432** of the end plate **430** to the lower end **304** of the charge tube **300**.
(52) FIG. **5**B is a second perspective view of the bottom end plate **430**. In this view, the proximal end **432** and distal end **434** of the carrier plate **430** are visible. Also shown is the closed end surface **435** and a central flange **436**. The central flange **436** receives the lowermost end **504** of the gun barrel housing **510**. The central flange **436** also receives bolt **820**. This may be used to secure the gun barrel housing **510** to the end plate **430**. Of interest, the ground pin **710** and electrical pins **720′**, **720″** are visible.
(53) Note that each of the electrical pins **720′**, **720″** extends into the bottom end plate **430**. As demonstrated in FIG. **8**, each pin **720′**, **720″** is received within a bulkhead **610**, **620**, respectively. Thus, end plate **430** contains two through-openings, each of which receives a bulkhead for securing an electrical pin. Because the pins **720′** and **720″** and their associated bulkheads **610**, **620**, are extremely small (certainly smaller than bulkhead **475** of FIG. **4**A), the bulkheads **610**, **620** may be referred to as “mini-bulkheads.”
(54) In the present disclosure, a unique “bridged” bulkhead assembly is optionally provided. The bridged bulkhead assembly provides an efficient way to install pre-wired pins into the carrier end plate **430** for field-connection with the addressable switch (shown at **760** in FIG. **8**).
(55) FIG. **6**A is a first perspective view of a bridged bulkhead assembly **600** of the present invention, in one embodiment. The bulkhead assembly **600** holds the signal transmission pin **720′** and the detonation pin **720″**.
(56) FIG. **6**B is a second perspective view of the bridged bulkhead assembly **600** of FIG. **6**A. Here, the view is seen from an end that is opposite the end of FIG. **6**A. Note that the bulkhead assembly **600** has also been flipped upside down relative to FIG. **6**A.
(57) In each of FIGS. **6**A and **6**B, a first bulkhead is shown at **610**, while a second bulkhead is shown at **620**. Bulkhead **610** has a first end **612**, and a second end **614** opposite the first end **612**. Similarly, bulkhead **620** has a first end **622**, and a second end **624** opposite the first end **622**. Bulkhead **610** is made up of body **615** while bulkhead **620** comprises body **625**. Each of bodies **615**, **625** is fabricated from an electrically non-conductive material. In one aspect, the bodies **615**, **625** are fabricated through an additive manufacturing process.
(58) Signal transmission line **410** feeds into the first end **612** of the first bulkhead **610**. The signal

transmission line **410** is securely connected to a first end of the signal transmission pin **720**′. This is seen more fully in the cross-sectional view of FIG. **7**, discussed below.
(59) In a similar way, a detonator wire **540** extends out from the first end **622** of the second bulkhead **620**. The detonator wire **540** is securely connected to a first end of the detonation pin **720**″. This is also shown more fully in the cross-sectional view of FIG. **7**. As discussed below, the detonator wire **540** may also be a first leg wire connected to the detonator **1000**, **1400**.
(60) Of interest, a second end **618** of the signal transmission pin **720**′ extends out from the second end **614** of the first bulkhead **610**. Similarly, a second end **628** of the detonation pin **720**″ extends out from the second end **624** of the second bulkhead **620**. Each of these second ends **618**, **628** represents a banana clip.
(61) FIGS. **6**A and **6**B also show a bridge **630**. The bridge **630** connects and also spaces apart the first **610** and second **620** bulkheads. Again, this is an optional arrangement for the detonator embodiments described herein.
(62) FIG. **7** is a cross-sectional view of the bridged bulkhead assembly **600** of FIGS. **6**A and **6**B. In this view, the signal transmission pin **720**′ is seen residing within the body **615** of the first bulkhead **610**. At the same time, the detonation pin **720**″ is seen residing within the body **625** of the second bulkhead **620**. Each of the signal transmission pin **720**′ and the detonation pin **720**″ is an electrically conductive pin. Preferably, each of the pins **720**′, **720**″ represents a single pin housed within a respective bulkhead **610**, **620**, that transmits electrical signals. The brass pins **720**′, **720**″ comprise a series of radial steps designed to maintain the pins **720**′, **720**″ within the respective bulkheads **610**, **620** to ensure a high-pressure barrier when an upstream perforating gun is detonated.
(63) It can be seen in FIG. **7** that the signal transmission line **410** is connected to the signal transmission pin **720**′ at the first end **612** of the first bulkhead **610**. At the same time, the detonator wire **540** is connected to the detonation pin **720**″ at the first end **622** of the second bulkhead **620**. In the present arrangement, the detonator wire **540** may not be used to connect to the detonator **1000**, **1400**.
(64) In a preferred arrangement, the body **615** of the first bulkhead **610** extends into a first opening of the carrier end plate **430**. At the same time, the body **625** of the second bulkhead **620** extends into a second opening of the end plate **430**. O-rings **650** encircle the bodies **615**, **625**, providing a seal within the openings of the end plate **430**. As noted in regard to FIGS. **4**A and **5**A, the end plate **430** is the carrier end plate that resides at a lower end **304** of the perforating gun charge tube **300**.
(65) Preferably, each bulkhead **610**, **620** includes compliant tabs. The tabs are seen partially at **425** in FIG. **4**A. The tabs **425** are configured to mate with slots **325** in the charge tube **300** at end **304**. This ensures a proper orientation of the pins **720**′, **720**″. Once the bulkheads **610**, **620** are installed, the bulkheads **610**, **620** are unable to back out of the end plate **430**. This removes the need for retainer nuts or other retention parts.
(66) The signal transmission pin **720**′ transmits electrical signals through the end plate **430** in a first direction. At the same time, the detonation pin **720**″ transmits the detonation signals back up through the end plate **430** in a second direction opposite the first direction. Preferably, the first direction is downstream while the second direction is upstream. The detonation pin **720**″ sends a detonation signal to a post **1030** or **1430** of a detonator **1000** or **1400**.
(67) FIG. **8** is a side, cross-sectional view of an explosive initiation assembly **800** of the present invention, in one embodiment. The explosive initiation assembly **800** is threadedly connected at opposing ends to gun barrel housings **510**, forming, for example, a part of the perforating gun assembly **200** of FIG. **2**.
(68) The explosive initiation assembly **800** first includes a tandem sub **700**. The tandem sub **700** represents a short tubular body having male threads at opposing ends **702**, **704**. Each opposing end **702**, **704** is connected to a gun barrel housing **510**. Intermediate the opposing ends **702**, **704** is a shoulder **706**. The gun barrel housings **510** are threaded onto the tandem sub **700** until the gun

barrel housings **510** engage with the shoulder **706**. Additional details concerning the tandem sub **700** are described in the parent application in connection with FIG. **4**.
(69) Residing within the tandem sub **700** is a switch housing **750**. Specifically, the switch housing **750** resides within a bore of the tandem sub **700**. A perspective view of the switch housing **750** is shown in FIG. **12** of the parent application.
(70) The explosive initiation assembly **800** also includes an addressable switch **760**. The addressable switch **760** resides within the switch housing **750**. A perspective view of the addressable switch **760** is shown in FIG. **13** of the parent application.
(71) The addressable switch **760** receives electrical signals sent by the operator from the surface **105**, through signal transmission pin **720**′, and filters those signals to identify an activation signal. If an activation signal is identified, then a detonation signal is separately sent for detonation of charges **320** in an adjacent (typically upstream) perforating gun **210**. The detonation signal may be sent through the detonation pin **720**″. Alternatively, where no detonation pin **720**″ is used, the detonation signal may be sent through a traditional leg wire that is connected to the detonator. In either instance, the detonation signal will energize an initiator, such as a fuse head.
(72) As seen in FIG. **8**, the transmission pin **720**′ and the detonation pin **720**″ extend through the bottom end plate **430** and into the switch housing **750**. Note that neither pin **720**′ nor **720**″ is at any time in electrical communication with the detonator **229**. Additional details of the switch housing **750** and the addressable switch **760** are also provided in the parent application in connection with FIGS. **12**, **13**, **16** and **17** and need not be repeated herein.
(73) The tandem sub **700** and its switch housing **750** reside between the bottom end plate **430** and the top end plate **420**. Flange members **436**, **426** associated with the bottom end plate **430** and the top end plate **420**, respectively, abut opposing ends of the tandem sub **700**. Beneficially, the end plates **430**, **420** mechanically seal the tandem sub **700**, protecting the addressable switch **760** from wellbore fluids and debris generated during detonation of the charges **320**.
(74) The bottom end plate **430** is shown herein in FIGS. **4**A, **5**A and **5**B, as discussed above. The top end plate **420** is shown in FIGS. **4**A, **4**B and **4**C of the present application and in FIG. **11**A of the parent application. As shown in FIG. **11**A, the top end plate **420** has a proximal end (shown in FIG. **11**A as **622**) and a distal end (shown in FIG. **11**A as **624**). Intermediate the proximal and distal ends is the flange **426**. As shown in FIG. **8** herein, the downstream end **704** of the tandem sub **700** abuts and shoulders-out against the flange **426**.
(75) The contact pin **470** resides within the non-conductive bulkhead **475**. A first (or proximal) end of the contact pin **470** extends into the switch housing **750** while a second (or distal) end of the contact pin **470** extends into the top end plate **420**. The contact pin **470** is used to transmit electrical signals through the tandem sub **700** down to the next perforating gun **210** via signal transmission line **410**, while the bulkhead **475** provides electrical insulation between the brass contact pin **470** and the surrounding metal tandem sub **700**.
(76) FIG. **9** is a perspective view of a detonator **1000** of the present invention, in one embodiment. Here, the detonator **1000** has been placed within the charge tube **300** for a perforating gun **210**. More specifically, the detonator **1000** has been snapped into place within a compartment **309**, much like, for example, a battery might be snapped into place within the inner housing of a flashlight.
(77) It can be seen that the charge tube **300** is connected to the bottom end plate **430**. The compliant tab **425** is seen having been received within slot **325**. The post **1030** of the detonator **1000** contacts the second end of the detonation pin **720**″. Sending an electrical current through the detonation pin **720**″ and into the detonator **1000** causes an explosive material to be heated. Alternatively, current may be sent directly from the addressable switch **760** through a first leg wire (shown at **1042** of FIG. **12**B and **1435** of FIG. **14**B as a resistor wire) and to the detonator **1000**.
(78) FIG. **10** is a perspective side view of the detonator **1000** of FIG. **9**. The detonator **1000** is configured to be snapped into a compartment **309** that resides within a charge tube **300**. The detonator **1000** has a lower end **1012** and an upper end **1014**.

(79) The detonator **1000** comprises a tubular body **1010**. Preferably, the tubular body **1010** is fabricated from aluminum. The tubular body **1010** includes a lower bore **1050** that houses an explosive material **1005**.
(80) Immediately above the tubular body **1010** is an insulating element **1020**. The insulating element **1020** defines a non-conductive (or electrically insulating) material such as rubber or synthetic rubber. The insulating element **1020** has a shoulder **1025**. The shoulder **1025** is configured to facilitate holding the detonator **1000** in place within the compartment **309** of the charge tube **300**.
(81) Above the insulating element **1020** is an upper conductive post **1030**. In one aspect, the upper conductive post **1030** is fabricated from brass and acts as a terminal for the detonator **1000**. The detonator **1000** and its conductive post **1030** function essentially like a AA battery or, perhaps, as a glass fuse in that they become part of an electrical circuit when snapped into place within the compartment **309**. The conductive post **1030** forms one leg of an initiator circuit while the tubular body **1010** forms another leg.
(82) In one aspect, a first leg wire (shown in FIG. **12**B at **1335**) is connected to the brass post **1030** while a second leg wire (seen in FIGS. **11**B and **12**B at **1046**) is wrapped around a lower end of the insulative element **1020**. The first leg wire **1335** may be run through an upper bore **1035** within the conductive post **1030**.
(83) In the preferred embodiment, when the detonator cartridge **1000** is snapped into place within the compartment **309** of the charge tube **300**, the conductive post **1030** resides near the proximal end **432** of the bottom end plate **430**. The conductive post **1030** is then in contact with the detonation pin **720**″ to receive current on the “hot voltage side.” Alternatively, the conductive post **1030** is in contact with the first leg wire **1335** coming off of the addressable switch **760** for delivery of current on the “hot voltage side.” Alternatively, a pair of contact terminals **722** (shown in FIG. **15**) may be provided for the hot voltage side. At the same time, the second leg wire **1046** or **1446** is in electrical communication with a conductive element (shown in FIG. **15** at **725**) on the ground side. The conductive elements **722**, **725** may be fabricated from stamped sheet metal.
(84) The detonator cartridge **1000** is designed to be inserted into the compartment **309** within the charge tube **300**. The compartment **309** contains the ground terminal **725**. The ground terminal **725**, in turn, is in electrical communication with a second leg wire extending from the addressable switch **760**. In this instance, the ground wire **1044** or **1444** is in electrical communication with the second leg wire **1046** or **1446** through the ground terminal **725** and through the tubular body **1010** or **1410** itself.
(85) In another instance, the second leg wire **1046** or **1446** of the addressable switch **760** is wrapped around or crimped to the tubular body **1010** or **1410** adjacent the ground wire **1044** or **1444**. In other words, the second leg wire may be connected to the tubular body **1010** or **1410** either directly or through the ground terminal **725**.
(86) In operation, the detonator **1000** may be held in place by means of clips **319**. The clips **319** are preferably fabricated from a non-conductive and pliable material such as a soft plastic. A separate conductor (conductive element shown in FIG. **15** at **725**) is provided adjacent the clips **319**. The conductive element **725** may be fabricated from stamped sheet metal, and serves as a ground terminal. In addition, radial terminals (conductive elements shown in FIG. **15** at **722**) may serve as hot leads. The terminals **722** are in contact with the conductive body **1410** and contact a conductive post **1430**.
(87) FIG. **11**A is a side, plan view of the detonator **1000** of FIG. **10**. The overall cartridge shape of the detonator **1000** can be seen.
(88) FIG. **11**B is a cross-sectional view of the detonator **1000** of FIG. **10**. The view is taken across Line A-A of FIG. **11**A. Here, an initiator **1205** is seen. The initiator **1205** comprises the first leg wire **1335**, a resistor wire **1042**, a resistor **1045**, a ground wire **1044**, and second leg wire **1046**. The upper bore **1035** extends from an opening **1040** at an upper end of the post **1030**. The upper bore

1035 is configured to closely receive the first leg wire **1335**, which is connected to the resistor wire **1042**. (Again, the first leg wire **1335** and the resistor wire **1042** may be the same wire, as an optional way of placing the resistor **1045** in electrical communication with the addressable switch **760**.)
(89) FIG. **12**A is another side, perspective view of the detonator **1000** of FIG. **10**. In this view, the upper conductive post **1030** has been removed, revealing a spring **1037**. The spring **1037** biases the upper post **1030** upward towards the bottom end plate **430**. In an alternative arrangement, the conductive post **1030** is not used and the spring **1037** itself becomes the terminal. In this instance, the spring **1037** is fabricated from a highly conductive material such as brass or copper and is connected to the resistor wire **1042**.
(90) FIG. **12**B is a cross-sectional view of the detonator **1000** of FIG. **12**A. The spring **1037** is used as the terminal in lieu of the post **1030**. The spring **1037** is in electrical communication with the resistor wire **1042**. The initiator **1205** is again indicated as a combination of parts **1335**, **1042**, **1044**, **1045**, and **1046**.
(91) An explosive material **1005** resides within the tubular body **1010** within the lower bore **1050**. The explosive material **1005** may be, for example, 1,000 mg of RDX material. In this embodiment, the resistor **1045** resides within or is in contact with the explosive material **1005**. For this reason, the resistor **1045** may be referred to as a dual-resistorized fuse head. In any instance, there is no need to connect the detonator cartridge **1000** with a wire in the field as the first **1335** and the second **1046** leg wires are pre-connected to the detonator **1000** in the shop. The detonator **1000** is then simply snapped into place within the compartment **309**.
(92) FIG. **13**A is a cross-sectional view of the detonator **1000** of FIG. **12**A. FIG. **13**B is a cut-away view of the detonator **1000** of FIG. **13**A. The upper housing **1030** has been installed in lieu of the spring **1037**. The post **1030** defines a terminal.
(93) It is observed that the resistor wire **1042** has been received within the opening **1040**. The resistor wire **1042** extends down to the resistor **1045**. From the resistor **1045**, a ground wire **1044** returns back up to the insulative material **1020**. The ground wire **1044** is crimped to an internal wall of the tubular body **1010**. The ground wire **1044** is in electrical communication with the second leg wire **1046** through the tubular body **1010** itself.
(94) FIG. **14**A is a plan view of a detonator **1400** in an alternate embodiment. FIG. **14**B is a cross-sectional view of the detonator **1400** of FIG. **14**A, taken across Line A-A of FIG. **14**A. The detonator **1400** will be discussed with reference to FIGS. **14**A and **14**B together.
(95) The detonator **1400** is generally in accordance with the detonator **1000** of FIGS. **10**, **11**A and **11**B. In this respect, the detonator **1400** comprises a tubular body **1410**. The detonator **1400** has a lower end **1412** and an upper end **1414**, with the tubular body **1010** being sealed at the lower end **1412**.
(96) Immediately above the tubular body **1410** is an insulating element **1420**. The insulating element **1420** defines a non-conductive (or electrically insulating) material such as rubber (including synthetic rubber). The insulating element **1420** has a shoulder **1425**, forming a flange. The flange **1425** may facilitate holding the detonator **1400** in place within the compartment **309** of the charge tube **300** as shown in FIG. **15**.
(97) Above the insulating element **1420** is an upper conductive housing, or post **1430**. When the detonator cartridge **1400** is snapped into place within the charge tube **300**, the post **1430** resides near the proximal end **432** of the bottom end plate **430**.
(98) Residing within the upper portion of the detonator **1400** is an initiator **1405**. The initiator **1405** is made up of two resistors **1445**, **1448** and a series of wires **1441**, **1442**, **1444** connecting the resistors **1445**, **1448** to the tubular body **1010**. In addition, the initiator **1405** includes a catalyst element **1443**.
(99) As shown in FIG. **14**B, a first leg wire **1435** is provided along the upper housing **1430**. The first leg wire **1435** extends from the addressable switch **760** and is crimped to the inner wall of the

post **1430**. Also seen is a resistor wire **1442**. The resistor wire **1442** is in electrical communication with the first leg wire **1435** by means of the conductive post **1430** itself. Of course, the first leg wire **1435** and the resistor wire **1442** may be the same wire.

(100) The resistor wire **1442** extends to a first resistor **1445**. Of interest, the resistor wire **1442** and first resistor **1445** reside entirely within the insulating element **1420**.

(101) A lower portion of the tubular body **1010** comprises a bore **1450**. The bore **1450** holds the explosive material **1005**. It should be noted that the resistor wire **1442** does not extend through the non-conductive (or electrically insulating) material such as rubber **1020**. Instead, the resistor wire **1442** is interrupted by the first resistor **1445**.

(102) Below the first resistor **1445** is a fuse wire **1441**. The fuse wire **1441** extends downward into the explosive material **1005**, then circles back up to the second resistor **1448**. A ground wire **1444** extends from the second resistor **1448** and crimps into the inner wall of the tubular body **1410**. The ground wire **1444** is in electrical communication with the second resistor leg **1446** by means of the tubular body **1410** itself.

(103) Intermediate the first **1445** and second **1448** resistors is a catalyst element **1443**. The catalyst element **1443** preferably represents a third resistor. In this arrangement, the catalyst element **1443** extends into the bore **1450** and is in contact with the explosive material **1005**. The catalyst element is a part of the initiator **1405**.

(104) In one aspect, the post **1430** does not serve an electrically conductive function. Instead, a conductive material is placed within the post **1430** in electrical communication with the resistor wire **1442**. The resistor wire **1442** extends into the insulating element **1420** to the first resistor **1445**. The first resistor **1445** resides entirely within the insulating element **1420**, but sends current along fuse wire **1441** to the catalyst element **1443**.

(105) FIG. **15** is a transparent view of the charge tube **300** of FIG. **9**. It can be seen that the cartridge that is the detonator **1400** has been snapped into a plastic receptacle within the charge tube **300**. In this arrangement, the receptacle represents a pair of elastomeric or plastic clips **319**. The detonator **1400** is adjacent to a detonator cord **910**.

(106) In a preferred embodiment, the detonator **1400** does not mate within the charge tube **300** via sliding engagement; rather, the detonator **1400** is mated into the plastic receptacle from above. The plastic receptacle, or compartment **309**, includes two stamped sheet-metal terminals. One terminal **725** acts as a ground for the detonator **1400**, while the other terminal **722** acts as a hot terminal, and has a crimp location to attach the first leg wire as the “hot” voltage side for the detonator **1400**.

(107) In operation, an electrical signal is sent from the surface **105** through the electric line **240**. The signal reaches the perforating gun assembly **200**. Typically, a lowest perforating gun **210** is designated for first explosive initiation. In that case, the signal passes along the internal signal transmission line **410** through each perforating gun **210** and is then passed along by the transmission pins **720′**, the addressable switches **760** in each tandem sub **225**, and the contact pins **470** until the signal reaches the lowest tandem sub **225** and its addressable switch **760**. The addressable switch **760** then recognizes the electrical signal and sends a detonation signal back up through the detonation pin **720″** and to the detonator **1000**.

(108) Upon reaching the detonator **1000**, current travels through the first leg wire **1435**, through the resistor wire **1442**, through the first resistor **1445**, along the fuse wire **1441**, through the catalyst **1443**, back up to the second resistor **1448**, through the ground wire **1444**, and to the second resistor leg **1446**.

(109) Based on the embodiments of a detonator **1000**, **1400** described above, a method of firing charges into a wellbore casing is also provided. In the method, the wellbore casing resides within the horizontal portion of a wellbore.

(110) FIG. **16** is a flow chart demonstrating steps for a method **1600** of firing shots into a wellbore casing. Of course, the shots will penetrate through the casing **150** and into the surrounding formation **115**.

(111) In one embodiment, the method **1600** first comprises providing a perforating gun assembly. This is shown in Box **1610** of FIG. **16**. The perforating gun assembly includes a gun barrel housing, and a charge tube residing within the gun barrel housing. The perforating gun assembly will also have a plurality of charges residing along the charge tube, as well as a detonator cord extending to each of the charges within the charge tube. Additionally, the perforating gun assembly will include an addressable switch as shown and discussed above in connection with FIG. **8**.
(112) The method **1600** also includes providing a detonator. This is seen at Box **1620**. The detonator may be in accordance with any of the detonator embodiments described above. In one aspect, the detonator comprises: a tubular body (such as tubular body **1410**) having an upper end and a lower end; a post (such as post **1430**) residing at the upper end; a lower bore (such as bore **1450**) within the tubular body housing an explosive charge material (**1005**) proximate the lower end (**1412**); and an initiator (such as initiator **1405** and components **1441**, **1442**, **1443**, **1444**, **1445**, **1448**) residing at least partially within the explosive material.
(113) Preferably, the post is fabricated from brass or copper while the tubular body is fabricated from aluminum.
(114) The addressable switch will have two so-called leg wires. In the method, these are first and second leg wires. In one embodiment, each leg wire extends to and is connected to the detonator. The first leg wire is crimped to (or is otherwise in contact with) the post while the second leg wire is crimped to (or is otherwise in contact with) the tubular body of the detonator below the post.
(115) In another embodiment, the first leg wire is crimped to a so-called hot terminal along a compartment of the charge tube, while the second leg wire is crimped to a ground terminal in the compartment. The crimping of the first and second leg wires may be done in the shop before the perforating gun assembly is taken to a well site.
(116) In one aspect, the detonator also includes an insulative element. The insulative element separates the terminal at the upper end of the cartridge from the lower bore within the tubular body. The insulative element may include: an upper end having an outer diameter configured to be received within the post, or residing immediately below the post; a lower end having an outer diameter configured to be received within an upper end of the tubular housing above the lower bore, or otherwise immediately above the lower bore; and a flange between the upper and lower ends.
(117) Together the post, the insulative material and the tubular body form a cartridge.
(118) In the method **1600**, the initiator may comprise: a resistive wire; a catalyst element residing within the lower bore and in contact with the resistive wire and the explosive charge material; a fuse wire connected to the catalyst element; and a ground wire extending from the catalyst element and grounded to an internal wall of the tubular body at the second leg wire.
(119) Optionally, the method **1600** may further include: wrapping the first leg wire around the upper end of the insulative material under the post (shown at Box **1630**); and wrapping the second leg wire around the lower end of the insulative material within the tubular body such that the ground wire and the second leg wire are in electrical communication with one another (shown at Box **1640**).
(120) The method **1600** also comprises placing the detonator into a compartment within the charge tube. This is indicated at Box **1650**. With the two leg wires coming off of the addressable switch already connected to the detonator in secure fashion, the operator in the field need only snap the detonator into the compartment in order to prepare the perforating gun assembly. Alternatively, the two leg wires are pre-crimped onto hot and ground terminals. Either way, no crimping of wires for the detonator need take place in the field.
(121) The method **1600** may also include: pumping the perforating gun assembly into the horizontal portion of the wellbore using an electric wireline (shown at Box **1660**); sending an electrical signal from a surface and down the electric wireline to the perforating gun assembly, wherein the signal reaches the addressable switch (shown at Box **1670**); and sending a detonation

signal from the addressable switch to the detonator, causing the detonator cord to be ignited (shown at Box **1680**).
(122) The result of sending the activation signal to the igniter is that a power charge is ignited in the charge tube **300**, causing the charges to fire into the wellbore casing. This is provided at Box **1690**.
(123) In addition to the method of perforating a casing provided above, a method of preparing a perforating gun assembly is also provided herein. The method first comprises providing a perforating gun assembly. In this case, the perforating gun assembly comprises a gun barrel housing, a charge tube residing within the gun barrel housing, a plurality of charges residing along the charge tube, a detonator cord extending to each of the charges within the charge tube, an addressable switch, and a compartment within the charge tube.
(124) The method also includes providing a detonator. The detonator comprises: a tubular body having an upper end and a lower end; a terminal residing at the upper end; a lower bore within the tubular body housing an explosive charge material proximate the lower end; and an initiator.
(125) The method additionally includes placing a first leg wire of an addressable switch in electrical communication with the terminal. The method then includes placing a second leg wire of the addressable switch in electrical communication with the tubular body.
(126) The method further comprises transporting the perforating gun assembly to a well site. After the perforating gun assembly has arrived at the well site, the method includes placing the detonator into the compartment within the charge tube.
(127) In one aspect of the method, the initiator comprises: a resistor wire in electrical communication with the first leg wire; a catalyst element residing within the lower bore and in contact with the explosive charge material; and a ground wire extending from the catalyst element and grounded to an internal wall of the tubular body.
(128) The second leg wire may be connected to the ground terminal, or it may be wrapped around the tubular body. Preferably, the compartment comprises a ground terminal in contact with the tubular body. Preferably, the detonator is placed into the compartment using a friction-fit or snap-fit arrangement. Placing the detonator into the compartment causes the ground terminal to contact the tubular body, and causes the post to contact a hot terminal.
(129) It should be understood that this description is not intended to limit the invention; on the contrary, the exemplary embodiments are intended to cover alternatives, modifications, and equivalents, which are included within the spirit and scope of the invention as defined by the appended claims. Further, in the detailed description of the exemplary embodiments, specific details are set forth in order to provide a comprehensive understanding of the claimed invention. However, one skilled in the art would understand that various embodiments may be practiced without such specific details.
(130) Further, variations of the detonator and of methods for using the detonator within a wellbore may fall within the spirit of the claims, below. It will be appreciated that the inventions are susceptible to other modifications, variations, and changes without departing from the spirit thereof.

Claims

1. A detonator for a perforating gun, the perforating gun having a charge tube holding a plurality of charges and a compartment, and the detonator comprising: a cartridge comprising: a tubular body having a first end, and a second end opposite the first end; a conductive post at the first end of the tubular body, with the conductive post being configured to be placed in electrical communication with an addressable switch along a perforating gun assembly; and a lower bore within the tubular body below the conductive post and housing an explosive charge material; an initiator residing within the cartridge, wherein the initiator comprises: a first resistor residing within the cartridge;

and a ground wire extending from the first resistor and connected to a wall of the tubular body; and wherein: the cartridge is configured to be placed within the compartment of the charge tube; the compartment comprises a hot terminal and a ground terminal; and placement of the cartridge into the compartment causes (i) the conductive post to be in contact with the hot terminal, (ii) the ground terminal to be in contact with the wall of the tubular body proximate the lower bore, and (ii) the conductive post to be in electrical communication with the addressable switch through the hot terminal.

2. The detonator of claim 1, wherein: the terminal and the tubular body are each fabricated from a metal.

3. The detonator of claim 2, wherein: the conductive post is fabricated from brass or copper; and the tubular body is fabricated from aluminum.

4. The detonator of claim 2, wherein: the perforating gun is adjacent a carrier end plate; the carrier end plate comprises a detonation pin in electrical communication with the addressable switch such that placement of the cartridge into the compartment causes the terminal to be in electrical communication with the addressable switch through the detonation pin.

5. The detonator of claim 2, wherein: the addressable switch has a first leg wire and a second leg wire, with the first leg wire being connected to the hot terminal, and the second leg wire being connected to the ground terminal.

6. The detonator of claim 2, further comprising: an insulative element separating the terminal at the upper end of the cartridge from the lower bore within the tubular body.

7. The detonator of claim 6, wherein: the insulative element is fabricated from molded rubber or synthetic rubber; and the detonator resides within the compartment of the charge tube adjacent a detonator cord.

8. The detonator of claim 7, wherein the insulative element further comprises a flange between upper and lower ends of the insulative element, with the flange being configured to seat within the compartment of the charge tube.

9. The detonator of claim 7, wherein: the first resistor is in contact with the explosive material and serves as a dual-resistorized fuse head; the addressable switch has a second leg wire in electrical communication with the ground wire; and the ground wire is in electrical communication with the second leg wire through the tubular body itself and through the ground terminal.

10. The detonator of claim 7, wherein: the first resistor resides along the insulative element; and the initiator further comprises: a second resistor residing within or below the insulative element but above the explosive material; a catalyst element residing between the first resistor and the second resistor, with the catalyst element being placed within the lower bore and being in contact with the explosive material; and a fuse wire between the first resistor and the second resistor, with the catalyst element residing along the fuse wire; and wherein the ground wire resides between the second resistor and the tubular body.

11. The detonator of claim 10, wherein: the post has an upper end and a lower end; and the insulative material comprises: an upper end having an outer diameter configured to be received within the lower end of the post; a lower end having an outer diameter configured to be received within the upper end of the tubular housing above the lower bore; and a flange between the upper and lower ends.

12. The detonator of claim 2, wherein the detonator is received into the compartment through a friction-fit or snap-fit.

13. A method of firing charges into a wellbore casing, the wellbore casing residing within the horizontal portion of a wellbore, and the method comprising: providing a perforating gun assembly, the perforating gun assembly comprising a gun barrel housing, a charge tube residing within the gun barrel housing, a plurality of charges residing along the charge tube, a detonator cord extending to each of the charges within the charge tube, a compartment within the charge tube and an addressable switch; providing a detonator comprising: a tubular body having an upper end and a

lower end, with the tubular body being fabricated from a conductive metal; a post residing at the upper end of the tubular body, with the post also being fabricated from a conductive metal; a lower bore within the tubular body housing an explosive charge material proximate the lower end of the tubular body; and an initiator in contact with the explosive material; and placing the detonator into the compartment within the charge tube, wherein: the compartment comprises a ground terminal; and placement of the detonator into the compartment (i) places the post in electrical communication with the addressable switch, and (ii) places the tubular body in contact with the ground terminal.
14. The method of claim 13, wherein the detonator is placed into the compartment using a friction-fit or snap-fit arrangement.
15. The method of claim 13, wherein the initiator comprises: a first resistor residing within the cartridge; and a ground wire extending from the first resistor and connected to a wall of the tubular body.
16. The method of claim 15, wherein: the post is fabricated from brass or copper; and the tubular body is fabricated from aluminum.
17. The method of claim 16, wherein: the post is in electrical communication with the first resistor by means of a resistor wire; the perforating gun is adjacent a carrier end plate; the carrier end plate comprises a detonation pin in electrical communication with the addressable switch; and placement of the cartridge into the compartment causes the terminal to be in contact with the addressable switch through the detonation pin.
18. The method of claim 16, further comprising: pumping the perforating gun assembly into the horizontal portion of the wellbore using an electric wireline; sending an electrical signal from a surface and down the electric wireline to the perforating gun assembly, wherein the signal reaches the addressable switch; and sending a detonation signal from the addressable switch, through the detonation pin, and to the detonator, causing the detonator cord to be ignited, and causing the plurality of charges to fire into the wellbore casing.
19. The method of claim 18, wherein: the terminal comprises a post; and a spring residing within the post, with the spring configured to bias the post outwardly and away from the tubular body.
20. The method of claim 19, wherein: the compartment comprises a hot terminal; and the addressable switch has a first leg wire and a second leg wire, with the first leg wire being connected to the hot terminal, and the second leg wire being connected to the ground terminal.
21. The method of claim 16, wherein the initiator further comprises: a first leg wire; a catalyst element residing within the lower bore and in contact with the explosive charge material; and a fuse wire placing the first leg wire in electrical communication with the catalyst element.
22. The method of claim 21, wherein: the initiator further comprises a second resistor; the catalyst element resides along the fuse wire between the first and the second resistor; and the ground wire extends from the second resistor and is grounded to an internal wall of the tubular body.
23. The method of claim 18, wherein: the addressable switch comprises a first leg wire and a second leg wire; the post is in electrical communication with the addressable switch by means of the first leg wire; and the second leg wire is wrapped around the tubular body such that the tubular body is grounded.
24. The method of claim 16, wherein: the detonator further comprises: an insulative element separating the terminal at the upper end of the cartridge from the lower bore within the tubular body, wherein the insulative element comprises: an upper end having an outer diameter configured to be received within the post; a lower end having an outer diameter configured to be received within an upper end of the tubular housing above the lower bore; and a flange between the upper and lower ends.
25. The method of claim 24, wherein: the cartridge further comprises a hot terminal; placement of the cartridge into the compartment causes (i) the conductive post to be in contact with the hot terminal, (ii) the ground terminal to be in contact with the wall of the tubular body proximate the lower bore, and (iii) the conductive post to be in electrical communication with the addressable

switch through the hot terminal; and together the post, the insulative element and the tubular body form the cartridge.

26. A method of preparing a perforating gun assembly, comprising: providing a perforating gun assembly, the perforating gun assembly comprising a gun barrel housing, a charge tube residing within the gun barrel housing, a plurality of charges residing along the charge tube, a detonator cord extending to each of the charges within the charge tube, an addressable switch, and a compartment within the charge tube; providing a detonator comprising: a tubular body having an upper end and a lower end; a conductive post residing at the upper end; a lower bore within the tubular body housing an explosive charge material proximate the lower end of the tubular body; and an initiator in contact with the explosive material; placing a first leg wire of an addressable switch in electrical communication with the terminal; placing a second leg wire of the addressable switch in electrical communication with the tubular body; transporting the perforating gun assembly to a well site; and after the perforating gun assembly has arrived at the well site, placing the detonator into the compartment within the charge tube.

27. The method of claim 26, wherein the detonator is placed into the compartment using a friction-fit or snap-fit arrangement.

28. The method of claim 27, wherein the initiator comprises: a resistor wire in electrical communication with the first leg wire; a catalyst element residing within the lower bore and in contact with the explosive charge material; and a ground wire extending from the catalyst element and grounded to an internal wall of the tubular body; and wherein the compartment comprises a ground terminal in contact with the tubular body.

29. The method of claim 28, wherein either the second leg wire is connected to the ground terminal, or the second leg wire is wrapped around the tubular body.

30. A method of preparing a perforating gun assembly, comprising: providing a perforating gun assembly, the perforating gun assembly comprising a gun barrel housing, a charge tube residing within the gun barrel housing, a plurality of charges residing along the charge tube, a detonator cord extending to each of the charges within the charge tube, an addressable switch, a detonation pin in electrical communication with the addressable switch, and a compartment within the charge tube, with the compartment having a ground terminal; providing a detonator comprising: a tubular body having an upper end and a lower end; a hot terminal residing at the upper end; a lower bore within the tubular body housing an explosive charge material proximate the lower end; and an initiator in contact with the explosive material; transporting the perforating gun assembly to a well site; and after the perforating gun assembly has arrived at the well site, placing the detonator into the compartment within the charge tube such that the post is in electrical contact with the addressable switch, and the tubular body is in electrical contact with the ground terminal.

31. The method of claim 30, wherein: the detonator is placed into the compartment using a friction-fit or snap-fit arrangement the hot terminal comprises at least one contact; and placing the detonator into the compartment places the post in physical contact with the hot terminal.

32. The method of claim 31, wherein the initiator comprises: a resistor wire in electrical communication with the terminal; a catalyst element residing within the lower bore and in contact with the explosive charge material; and a ground wire extending from the catalyst element and grounded to an internal wall of the tubular body; and wherein the compartment comprises a ground terminal in contact with the tubular body.